

CS201 Composite

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Abstract—Write only ONE function per question in C++. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can not modify the input.

I. QUESTION

Given a directed graph represented by adjacency matrix, write a recursive method

```
bool hasCycle(int v, bool* visited)
```

in the Graph class that determines whether the graph contains any cycles starting from vertex v . Use the same idea in Depth-First Search (DFS). Your method should return true if there is at least one cycle in the graph, and false otherwise.

II. QUESTION

Write a function

```
LinkedList windowedPairwiseGCD(int k)
```

that computes the GCD (Greatest Common Divisor) of every window of k consecutive nodes in the list and returns a new linked list containing these GCD values.

Sample 1:

list: 18 → 24 → 30 → 42 → 54

GCDs in window size 3: 6 → 6 → 6

Sample 2:

list: 15 → 18 → 36 → 48 → 60

GCDs in window size 4: 3 → 6

III. QUESTION

Write a nonrecursive method using Stack

```
int leftistOrRightist()
```

that finds the difference between the number of leftist nodes and rightist nodes in binary search tree. A node is leftist (rightist) if it has only left (right) child.

IV. QUESTION

Given an array A of N integers, write a function

```
void topTwoFrequent(int* A, int N)
```

Your function should display the two most frequent elements in the given array A , which is already sorted. Your function must run in $O(N)$ time. You can not modify the input array A .

V. QUESTION

```

1 int algorithm1(int N){
2     int i, j, k, sum = 0;
3     for (i = 1; i <= N - 2 * i; i++)
4         sum++;
5     for (i = 1; i <= N; i++)
6         for (j = N; j >= 1; j /= 3)
7             sum++;
8     for (i = 0; i <= N; i++)
9         for (j = i + 1; j <= N * N; j++)
10            sum++;
11     return sum;
12 }
```

What does the function algorithm1 return in terms of N ? Assume that N is a power of 3.